ANSWER KEY & MARKING SCHEME SET 1

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 36 cm	$\sqrt{l^2+b^2+h^2}=\sqrt{144+576+576}$.
2	**(d)** 22176 cm^3	Volume of hemisphere + volume of cone.
3	**(d)** 125	$\frac{(5)^3}{1^3}=5^3$.
4	**(c)** 144	$\frac{54\times12\times6}{27}$.
5	**(d)** 45	Cylinder volume ÷ cone volume $=\frac{3h}{H}$.
6	**(a)** 2425.5 cm^3	$\frac{2}{3}\pi r^3$.
7	**(d)** 43659 cm^3	$\frac{1}{3}\pi r^2h$ with the given values.
8	**(b)** 146300 cm^3	πr^2h with $r=35$, $h=38$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$5^3=125$, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	TSA $=6a^2=96$, so A is true; R is true but the TSA is $6a^2$, so R does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Diagonal $=\sqrt{6^2+18^2+27^2}=\sqrt{1089}=33$ cm; TSA $=2(108+486+162)=1512$ cm^2 (1); Volume $=6\times18\times27=2916$ cm^3 (1).	1+1
12	Slant height of the cone $=\sqrt{r^2+h^2}=70$ cm; Volume $=\frac{2}{3}\pi r^3+\frac{1}{3}\pi r^2h=258720$ cm^3 (1); Surface area $=2\pi r^2+\pi rl=20328$ cm^2 (1).	1+1
13	$\pi r^2h=5544$ gives $r^2=\frac{5544\times7}{22\times4}=441$; $r=21$ cm (1); CSA $=2\pi rh=2\times\frac{22}{7}\times21\times4=528$ cm^2 (1).	1+1
14	Diagonal $=\sqrt{4^2+8^2+8^2}=\sqrt{144}=12$ cm; TSA $=2(32+64+32)=256$ cm^2 (1); Volume $=4\times8\times8=256$ cm^3 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{4}{3}\pi12^3=\pi(3)^2h$ (1); $h=\frac{4\times1728}{3\times9}=256$ cm (1); CSA $=2\pi rh=2\times\frac{22}{7}\times3\times256=4827.43$ cm^2 (1).	1+1+1
16	Volume of water $=\pi r^2h=\frac{22}{7}\times21^2\times15=20790$ cm^3 (1); Base area $=33\times14=462$ cm^2 (1); Height $=\frac{20790}{462}=45$ cm (1).	1+1+1
17	Capacity $=\frac{2}{3}\pi r^3=\frac{2}{3}\times\frac{22}{7}\times21^3=19404$ cm^3 (1); Number of bowls $=\frac{58212}{19404}=3$ (2).	1+2
	OR: $\frac{4}{3}\pi6^3=\pi(3)^2h$ (1); $h=\frac{4\times216}{3\times9}=32$ cm (1); CSA $=2\pi rh=2\times\frac{22}{7}\times3\times32=603.43$ cm^2 (1).	1+1+1
18	Volume of water $=\pi r^2h=\frac{22}{7}\times14^2\times20=12320$ cm^3 (1); Base area $=22\times14=308$ cm^2 (1); Height $=\frac{12320}{308}=40$ cm (1).	1+1+1
	OR: Capacity $=\frac{2}{3}\pi r^3=\frac{2}{3}\times\frac{22}{7}\times10.5^3=2425.5$ cm^3 (1); Number of bowls $=\frac{14553}{2425.5}=6$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $5\times4\times3=60$ m^3 (1). (ii) $60\times1000=60000$ litres (1). (iii) $2(20+12+15)=94$ m^2 (1). (iv) Open tank: $94-20=74$ m^2 (1); cost $=74\times15=$ Rs 1110 (1).	1+1+1+2
20	Volume of the cone $=\frac{1}{3}\pi r^2h=12936$ cm^3 (1); Volume of the hemisphere $=\frac{2}{3}\pi r^3=19404$ cm^3 (1); Volume per cone $=32340$ cm^3 (1); For 31 cones: $31\times32340=1002540$ cm^3 (2).	1+1+1+2
	OR: Volume of sphere $=\frac{4}{3}\pi6^3$ (1); Each cylinder has volume $\frac{4\pi216}{3\times5}=\pi(3)^2h$ (2); $h=\frac{4\times216}{3\times5\times9}=6.4$ cm (2).	1+2+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 125	$\frac{(20)^3}{4^3} = 5^3$.
2	**(c)** 672	$\frac{36 \times 24 \times 21}{27}$.
3	**(d)** 18	Cylinder volume ÷ cone volume $= \frac{3h}{H}$.
4	**(c)** 19404 cm^3	$\frac{2}{3}\pi r^3$.
5	**(a)** 9856 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
6	**(a)** 66528 cm^3	$\pi r^2 h$ with $r = 21$, $h = 48$.
7	**(d)** 30 cm	$\sqrt{l^2 + b^2 + h^2} = \sqrt{16 + 400 + 484}$.
8	**(c)** 26334 cm^3	Volume of hemisphere + volume of cone.
9	**(c)** A is true but R is false.	A is correct. R is false because the TSA is $2\pi rh + 2\pi r^2 = 2\pi r(r + h)$ (two circular ends).
10	**(d)** A is false but R is true.	A is false because the volume of a cone is $\frac{1}{3}\pi r^2 h$; R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Slant height of the cone $= \sqrt{r^2 + h^2} = 17.5$ cm; Volume $= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = 4042.5$ cm^3 (1); Surface area $= 2\pi r^2 + \pi rl = 1270.5$ cm^2 (1).	1+1
12	$\pi r^2 h = 1540$ gives $r^2 = \frac{1540 \times 7}{22 \times 10} = 49$; $r = 7$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 10 = 440$ cm^2 (1).	1+1
13	Diagonal $= \sqrt{6^2 + 6^2 + 7^2} = \sqrt{121} = 11$ cm; TSA $= 2(36 + 42 + 42) = 240$ cm^2 (1); Volume $= 6 \times 6 \times 7 = 252$ cm^3 (1).	1+1
14	Slant height of the cone $= \sqrt{r^2 + h^2} = 52.5$ cm; Volume $= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = 109147.5$ cm^3 (1); Surface area $= 2\pi r^2 + \pi rl = 11434.5$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{4}{3}\pi 9^3 = \pi(6)^2 h$ (1); $h = \frac{4 \times 729}{3 \times 36} = 27$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 6 \times 27 = 1018.29$ cm^2 (1).	1+1+1
16	Volume of water $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 30 = 18480$ cm^3 (1); Base area $= 44 \times 21 = 924$ cm^2 (1); Height $= \frac{18480}{924} = 20$ cm (1).	1+1+1
17	Capacity $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 10.5^3 = 2425.5$ cm^3 (1); Number of bowls $= \frac{4851}{2425.5} = 2$ (2).	1+2
	OR: $\frac{4}{3}\pi 9^3 = \pi(2)^2 h$ (1); $h = \frac{4 \times 729}{3 \times 4} = 243$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 2 \times 243 = 3054.86$ cm^2 (1).	1+1+1
18	Volume of water $= \pi r^2 h = \frac{22}{7} \times 21^2 \times 20 = 27720$ cm^3 (1); Base area $= 22 \times 21 = 462$ cm^2 (1); Height $= \frac{27720}{462} = 60$ cm (1).	1+1+1
	OR: Capacity $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 10.5^3 = 2425.5$ cm^3 (1); Number of bowls $= \frac{12127.5}{2425.5} = 5$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{22}{7} \times 49 \times 20 = 3080$ cm^3 (1). (ii) $2 \times \frac{22}{7} \times 7 \times 20 = 880$ cm^2 (1). (iii) $880 + 2 \times 154 = 1188$ cm^2 (1). (iv) 30.8 L $= 30800$ cm^3 (1); number of bottles $= \frac{30800}{3080} = 10$ (1).	1+1+1+2
20	Volume of the cone $= \frac{1}{3}\pi r^2 h = 12936$ cm^3 (1); Volume of the hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm^3 (1); Volume per cone $= 32340$ cm^3 (1); For 20 cones: $20 \times 32340 = 646800$ cm^3 (2).	1+1+1+2
	OR: Volume of sphere $= \frac{4}{3}\pi 9^3$ (1); Each cylinder has volume $\frac{4\pi 729}{3 \times 5} = \pi(1)^2 h$ (2); $h = \frac{4 \times 729}{3 \times 5 \times 1} = 194.4$ cm (2).	1+2+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 14: **Surface Area and Volume** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 9	Cylinder volume ÷ cone volume $= \frac{3h}{H}$.
2	**(a)** 64 cm	Equal volumes: $\frac{4}{3}\pi R^3 = \pi r^2 h$.
3	**(d)** 1617 cm^3	$\frac{1}{3}\pi r^2 h$ with the given values.
4	**(b)** 26488 cm^3	$\pi r^2 h$ with $r = 14$, $h = 43$.
5	**(b)** 26 cm	$\sqrt{l^2 + b^2 + h^2} = \sqrt{36 + 64 + 576}$.
6	**(a)** 23562 cm^3	Volume of hemisphere + volume of cone.
7	**(d)** 27	$\frac{(3)^3}{1^3} = 3^3$.
8	**(c)** 980	$\frac{175 \times 35 \times 20}{125}$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{4}{3} \times \frac{22}{7} \times 343 = \frac{4312}{3}$, using R.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Melting does not change the amount of metal, so A is true; R is a true formula but is not the reason for A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\pi r^2 h = 1848$ gives $r^2 = \frac{1848 \times 7}{22 \times 12} = 49$; $r = 7$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 12 = 528$ cm^2 (1).	1+1
12	Diagonal $= \sqrt{12^2 + 24^2 + 24^2} = \sqrt{1296} = 36$ cm; TSA $= 2(288 + 576 + 288) = 2304$ cm^2 (1); Volume $= 12 \times 24 \times 24 = 6912$ cm^3 (1).	1+1
13	Slant height of the cone $= \sqrt{r^2 + h^2} = 35$ cm; Volume $= \frac{2}{3}\pi r^3 + \frac{1}{3}\pi r^2 h = 32340$ cm^3 (1); Surface area $= 2\pi r^2 + \pi rl = 5082$ cm^2 (1).	1+1
14	$\pi r^2 h = 13552$ gives $r^2 = \frac{13552 \times 7}{22 \times 22} = 196$; $r = 14$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 14 \times 22 = 1936$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{4}{3}\pi 12^3 = \pi(4)^2 h$ (1); $h = \frac{4 \times 1728}{3 \times 16} = 144$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 4 \times 144 = 3620.57$ cm^2 (1).	1+1+1
16	Volume of water $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 10 = 6160$ cm^3 (1); Base area $= 11 \times 7 = 77$ cm^2 (1); Height $= \frac{6160}{77} = 80$ cm (1).	1+1+1
17	Capacity $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 10.5^3 = 2425.5$ cm^3 (1); Number of bowls $= \frac{7276.5}{2425.5} = 3$ (2).	1+2
	OR: $\frac{4}{3}\pi 3^3 = \pi(6)^2 h$ (1); $h = \frac{4 \times 27}{3 \times 36} = 1$ cm (1); CSA $= 2\pi rh = 2 \times \frac{22}{7} \times 6 \times 1 = 37.71$ cm^2 (1).	1+1+1
18	Volume of water $= \pi r^2 h = \frac{22}{7} \times 14^2 \times 15 = 9240$ cm^3 (1); Base area $= 22 \times 14 = 308$ cm^2 (1); Height $= \frac{9240}{308} = 30$ cm (1).	1+1+1
	OR: Capacity $= \frac{2}{3}\pi r^3 = \frac{2}{3} \times \frac{22}{7} \times 21^3 = 19404$ cm^3 (1); Number of bowls $= \frac{38808}{19404} = 2$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $l = \sqrt{7^2 + 24^2} = 25$ cm (1). (ii) $\pi rl = \frac{22}{7} \times 7 \times 25 = 550$ cm^2 (1). (iii) $\frac{1}{3} \times \frac{22}{7} \times 49 \times 24 = 1232$ cm^3 (1). (iv) Volume is unchanged: $\frac{22}{7} \times 49 \times h = 1232$ (1), so $h = 8$ cm (1).	1+1+1+2
20	Volume of the cone $= \frac{1}{3}\pi r^2 h = 103488$ cm^3 (1); Volume of the hemisphere $= \frac{2}{3}\pi r^3 = 155232$ cm^3 (1); Volume per cone $= 258720$ cm^3 (1); For 36 cones: $36 \times 258720 = 9313920$ cm^3 (2).	1+1+1+2
	OR: Volume of sphere $= \frac{4}{3}\pi 6^3$ (1); Each cylinder has volume $\frac{4\pi 216}{3 \times 5} = \pi(2)^2 h$ (2); $h = \frac{4 \times 216}{3 \times 5 \times 4} = 14.4$ cm (2).	1+2+2

Mistakes to avoid:

- Forgetting the factor $\frac{1}{3}$ in the volume of a cone or pyramid.
- Using the vertical height h instead of the slant height l in πrl (CSA of a cone).
- Writing the TSA of a cylinder as $2\pi rh + \pi r^2$; there are two circular ends, so it is $2\pi rh + 2\pi r^2$.
- Using the diameter as the radius, or leaving r in the formula when the problem gives d.
- Mixing units: converting cm^3 to litres needs $\div 1000$, and m^3 to litres needs $\times 1000$.
- For a hemisphere, using the sphere formulas without halving the volume, or taking TSA as $2\pi r^2$ and forgetting the flat circle ($3\pi r^2$ in all).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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